SQL Project - Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

Question Set 1 - Easy

1. What is the total number of records in the dataset?

```
SELECT COUNT(*) AS total_rows FROM nassau_candy_clean;
```

2. What does the raw data look like?

```
SELECT * FROM nassau_candy_clean LIMIT 10;
```

3. Are there any missing or invalid financial values?

```
SELECT *
FROM nassau_candy_clean
WHERE Sales IS NULL
OR Units IS NULL
OR Gross_profit IS NULL
OR Cost IS NULL
OR Sales <= 0
OR Units <= 0;
```

4. Which products generate the highest sales revenue?

```
SELECT
Product_Name,
SUM(Sales) AS total_sales
FROM nassau_candy_clean
GROUP BY Product_Name
ORDER BY total_sales DESC
LIMIT 10;
```

5. Which products generate the highest profit?

```
SELECT
Product_Name,
SUM(Gross_profit) AS total_profit
FROM nassau_candy_clean
GROUP BY Product_Name
ORDER BY total_profit DESC
LIMIT 10;
```

6. What is the overall average profit margin?

```
SELECT
ROUND(SUM(Gross_profit)/SUM(Sales)*100,2) AS overall_margin
FROM nassau_candy_clean;
```

7. What is the total cost incurred by the business?

```
SELECT
SUM(Cost) AS total_cost
FROM nassau_candy_clean;
```

8. What is the average profit per order?

```
SELECT
ROUND(SUM(Gross_profit) / COUNT(Order_ID), 2) AS avg_profit_per_order
FROM nassau_candy_clean;
```

Question Set 2 - Moderate

1. Which products have high sales but low margin?

```
SELECT *
FROM product_metrics
WHERE gross_margin < 15
ORDER BY total_sales DESC;
```

2. Which products should be discontinued due to low performance?

```
SELECT *
FROM product_metrics
WHERE total_sales < 5000
AND total_profit < 1000;
```

3. How are products distributed across margin categories?

```
SELECT
CASE
  WHEN gross_margin >= 30 THEN 'High Margin'
  WHEN gross_margin BETWEEN 15 AND 30 THEN 'Medium Margin'
  ELSE 'Low Margin'
END AS margin_category,
COUNT(*) AS product_count
FROM product_metrics
GROUP BY margin_category;
```

4. Which divisions generate the most revenue and profit?

```
SELECT
Division,
SUM(Sales) AS revenue,
SUM(Gross_profit) AS profit,
ROUND(SUM(Gross_profit)/SUM(Sales)*100,2) AS margin
```

```
FROM nassau_candy_clean
GROUP BY Division
ORDER BY revenue DESC;
```

5. How does cost impact product profitability?

```
SELECT
Product_Name,
SUM(Sales) AS sales,
SUM(Cost) AS cost,
SUM(Gross_profit) AS profit
FROM nassau_candy_clean
GROUP BY Product_Name
ORDER BY cost DESC;
```

6. Which regions are most profitable?

```
SELECT
Region,
SUM(Sales) AS revenue,
SUM(Gross_profit) AS profit,
ROUND(SUM(Gross_profit)/SUM(Sales)*100,2) AS margin
FROM nassau_candy_clean
GROUP BY Region
ORDER BY margin DESC;
```

7. What is the average order value by region?

```
SELECT
Region,
ROUND(SUM(Sales)/COUNT(Order_ID),2) AS avg_order_value
FROM nassau_candy_clean
GROUP BY Region;
```

8. Which products have high sales but low profit per unit?

```
SELECT
Product_Name,
SUM(Sales) AS total_sales,
SUM(Units) AS total_units,
ROUND(SUM(Gross_profit)/NULLIF(SUM(Units),0),2) AS profit_per_unit
FROM nassau_candy_clean
GROUP BY Product_Name
HAVING profit_per_unit < 1
ORDER BY total_sales DESC;
```

9. Which divisions have the highest cost-to-revenue ratio?

```
SELECT
Division,
SUM(Cost) AS total_cost,
SUM(Sales) AS total_sales,
ROUND(SUM(Cost)/NULLIF(SUM(Sales),0)*100,2) AS cost_ratio
FROM nassau_candy_clean
GROUP BY Division
ORDER BY cost_ratio DESC;
```

Question Set 3 – Advanced

1. Does the business follow the Pareto principle (80/20 rule)?

```
WITH product_profit AS (
  SELECT Product_Name, SUM(Gross_profit) AS profit
  FROM nassau_candy_clean
  GROUP BY Product_Name
),
ranked AS (
```

```
  SELECT *,
  SUM(profit) OVER () AS total_profit,
  SUM(profit) OVER (ORDER BY profit DESC) AS cumulative_profit
  FROM product_profit
)
SELECT *,
ROUND(cumulative_profit/total_profit*100,2) AS cumulative_percentage
FROM ranked;
```

2. Which products are destroying profitability?

```
SELECT *
FROM product_metrics
WHERE gross_margin < 15
AND total_sales > 1000;
```

3. How does product performance vary across regions?

```
SELECT
Region,
Product_Name,
SUM(Sales) AS sales,
SUM(Gross_profit) AS profit,
ROUND(SUM(Gross_profit)/SUM(Sales)*100,2) AS margin
FROM nassau_candy_clean
GROUP BY Region, Product_Name
ORDER BY Region, profit DESC;
```

4. Which shipping modes impact profitability the most?

```
SELECT
Ship_mode,
SUM(Sales) AS revenue,
SUM(Gross_profit) AS profit,
```

```
ROUND(SUM(Gross_profit)/SUM(Sales)*100,2) AS margin
FROM nassau_candy_clean
GROUP BY Ship_mode
ORDER BY margin DESC;
```

5. Which states contribute the most profit?

```
SELECT
State,
SUM(Sales) AS revenue,
SUM(Gross_profit) AS profit
FROM nassau_candy_clean
GROUP BY State
ORDER BY profit DESC;
```

6. Which products fall into high-risk category?

```
SELECT *,
CASE
  WHEN gross_margin < 20 THEN 'High Risk'
  WHEN gross_margin BETWEEN 20 AND 40 THEN 'Moderate Risk'
  ELSE 'Safe'
END AS risk_category
FROM product_metrics;
```

7. Which regions have high revenue but low profitability (hidden inefficiency)?

```
SELECT
Region,
SUM(Sales) AS revenue,
SUM(Gross_profit) AS profit,
ROUND(SUM(Gross_profit)/NULLIF(SUM(Sales),0)*100,2) AS margin
FROM nassau_candy_clean
GROUP BY Region
```

```
HAVING margin < 15
ORDER BY revenue DESC;
```

8. What is the total profit leakage due to low-margin products?

```
SELECT
SUM(Gross_profit) AS low_margin_profit
FROM nassau_candy_clean
WHERE Gross_profit / Sales < 0.15;
```

9. What percentage of total profit comes from top 10 products?

```
WITH top_products AS (
  SELECT
  Product_Name,
  SUM(Gross_profit) AS profit
  FROM nassau_candy_clean
  GROUP BY Product_Name
  ORDER BY profit DESC
  LIMIT 10
)
SELECT
ROUND(
  SUM(profit) * 100 /
  (SELECT SUM(Gross_profit) FROM nassau_candy_clean), 2) AS top_10_profit_contribution
FROM top_products;
```